CLIFF

CLIFF-LMD
CLIFF-RMD
CLIFF_LED_EN
CLIFF-RRD
CLIFF-RFD
CLIFF-LRD
CLIFF-LFD

CLIFF-LM
CLIFF-RM

File: CLIFF.kicad_sch

BUMPER

BUMP-L
BUMP-R
BUMP_L_LED_EN
BUMP_R_LED_EN

File: bumper_sensors.kicad_sch

SIDE SENSORS

VSENS-3V3
SIDE-R1
ZX-SH1_0-7PWT

USART2-IR-R-RX
SIDE-PROXI-CE-R
I2C4_SCL
I2C4_SDA

CARPET SENSOR

LO
ECHO_ADCD

CARPET-SENSOR-LOD
CARPET-SENSOR-ECHO

File: Carpet-sensor.kicad_sch

MAIN BRUSH MOTOR

MAIN-BRUSH-IN1
MAIN-BRUSH-IN2
MAIN-BRUSH-ADCD

File: MAIN-BRUSH-MOTOR .kicad_sch

WATCHDOG

WDI
WD_OKD

File: WATCHDOG.kicad_sch

WHEEL MOTORs

WHEEL-MOT-R-IN2
WHEEL-MOT-R-IN1
WHEEL-MOT-R-ENCODED
WHEEL-MOT-R-DROPD
WHEEL-MOT-L-IN2
WHEEL-MOT-L-IN1
WHEEL-MOT-L-DROPD
WHEEL-MOT-L-ENCODED
WHEEL-MOT-R-ADCD
WHEEL-MOT-L-ADCD

File: WHEEL-MOTORs .kicad_sch

LiDAR

WD_OK
LiDAR-RXD
LiDAR-TXD
LIDAR-MOTOR-CTRL
LiDAR-EN

LIDAR_RXD
LIDAR_TX
LIDAR-MOTOR-CTRLD

File: LIDAR.kicad_sch

MAIN FAN

MAIN-FAN-FBD
MAIN-FAN-PWM

File: MAIN-FAN .kicad_sch

USB-C POWER

DP
DN
USB_CC1
USB_CC2
USB_VBUS_SENSE
USB_PWR_EN

USB_CC1D
USB_CC2
USB_VBUS_SENSE
USB_PWR_END

File: usb_power.kicad_sch

SIDE BRUSH MOTORs

SIDE-BRUSH-IN1
SIDE-BRUSH-IN2
SIDE-BRUSH-ADCD

File: SIDE-BRUSH-MOTORs .kicad_sch

MCU-STM32

MAIN-BRUSH-IN1
MAIN-BRUSH-IN2
MAIN-BRUSH-ADC
WHEEL-MOT-R-IN2
WHEEL-MOT-R-IN1
WHEEL-MOT-R-ENCODE
WHEEL-MOT-R-DROP
WHEEL-MOT-L-IN2
WHEEL-MOT-L-IN1
WHEEL-MOT-L-DROP
WHEEL-MOT-L-ENCODE
WHEEL-MOT-R-ADC
WHEEL-MOT-L-ADC
MAIN-FAN-FB
MAIN-FAN-PWM
SIDE-BRUSH-IN1
SIDE-BRUSH-IN2
SIDE-BRUSH-ADC
HALL-BIN
HALL-TOWER
REAR-IR-L
REAR-IR-R
IMU-INT1
LED-HOME
LED-PWR
CLEAN-BTN
HOME-BTN
BAT_ID
DC_IN_SENSE
BAT_SENSE
VSENS-5V-USB-EN
VSENS_3V3_EN
PWR-BTN-SENSE
STM-PWR-CTRL
CARPET-SENSOR-LOD
CARPET-SENSOR-ECHO
INT-SENSORS
WDID
LIDAR_RXD
LIDAR_TX
LIDAR-MOTOR-CTRLD
DP
DN
USB_CC1D
USB_CC2
USB_VBUS_SENSE
USB_PWR_END
VMOT-14V4-END
VL-I2C-RESETD
UART3_RX2
VL-L-LPnD
UART3_RX1
WD_OK
PWR-SW-G
BAT_SENSE
DC_IN_SENSE

File: STM32G473.kicad_sch

VL-I2C-RESET
IR-F-R
VL-L-LPn
IR-F-L

FRONT SENSORS

VSENS-3V3
VL-R1
2005FV-2X4P
VL-R-LPn
R196
47k
VL-L1
2005FV-2X4P
VL-L-LPn
C6
10uF/10V
C19
10uF/10V

HALL-BIN1 / HALL-TOWER1 / REAR-IR1

HALL-BIN1
HALL-TOWER1
ZX-SH1_0-3PLT
REAR-IR1
ZX-SH1_0-4PLT

IMU

INT1D
SDA
SCL

File: IMU-ICM-4267-P .kicad_sch

BUTTON-LEDs

POWER-BTND
LED-HOME
LED-PWR
CLEAN-BTND
HOME-BTND

File: BUTTON-LEDs .kicad_sch

POWER CONTROL

VMOT-14V4-EN
STM-PWR-CTRL
PWR-BTN-SENSE
VSENS_3V3_EN
VSENS-5V-USB-EN
WD_OK
PWR-SW-G
POWER-BTN

File: SYSTEM-POWER.kicad_sch

BATTERY CHARGER

BATTERY-CHARGER
PWR-SW-G
BAT_SENSE
DC_IN_SENSE
ID

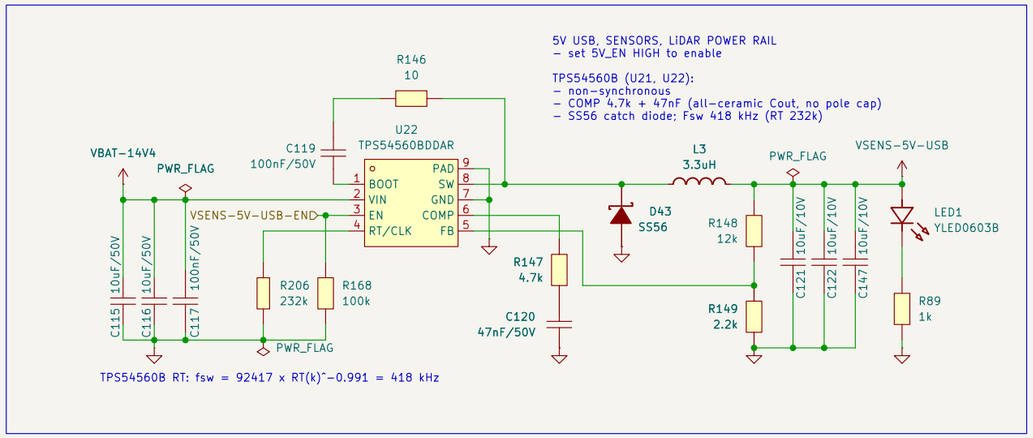
File: Battery-Charger.kicad_sch

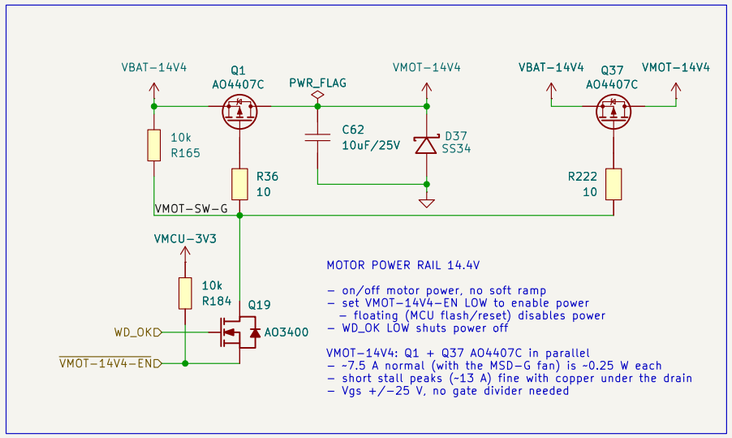
R2 47k
R6 2.2k
R5 2.2k

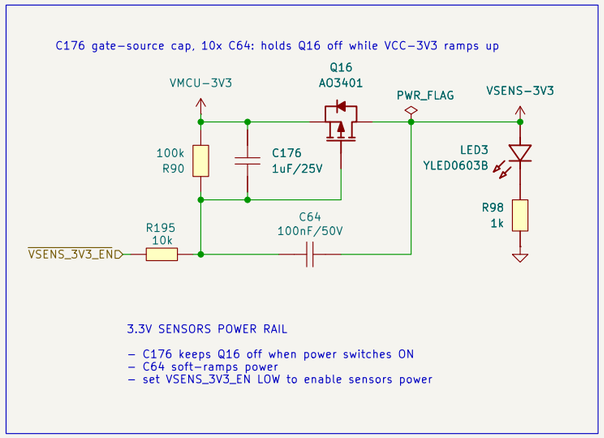
Sheet: /
File: Main.kicad_sch

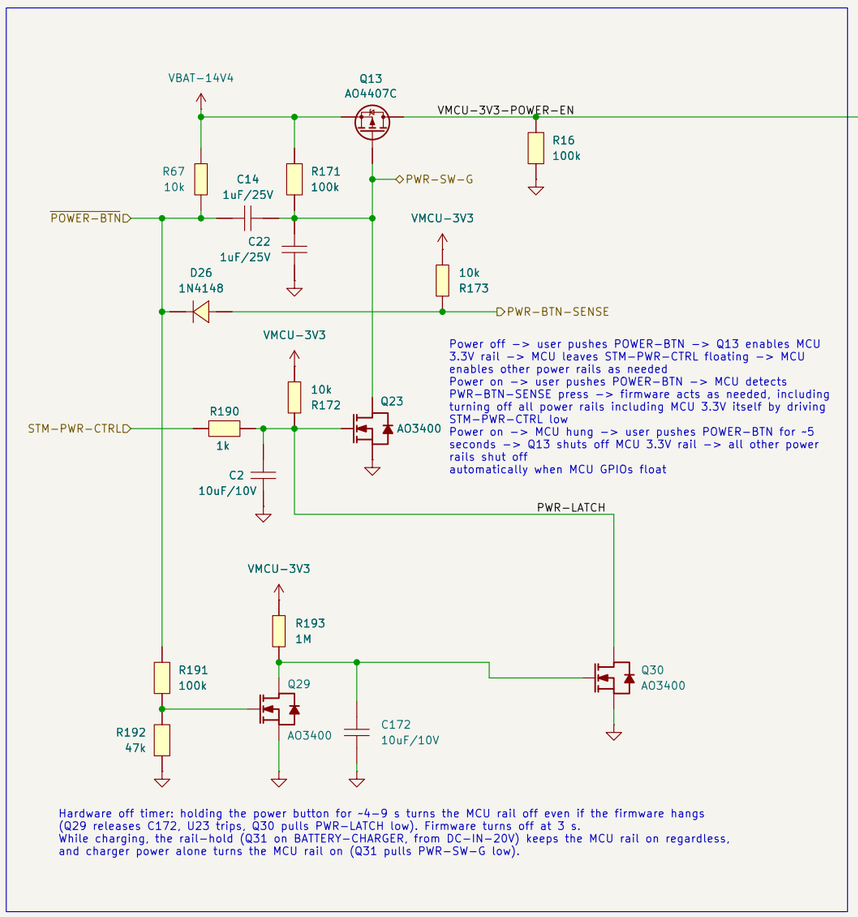
Title:

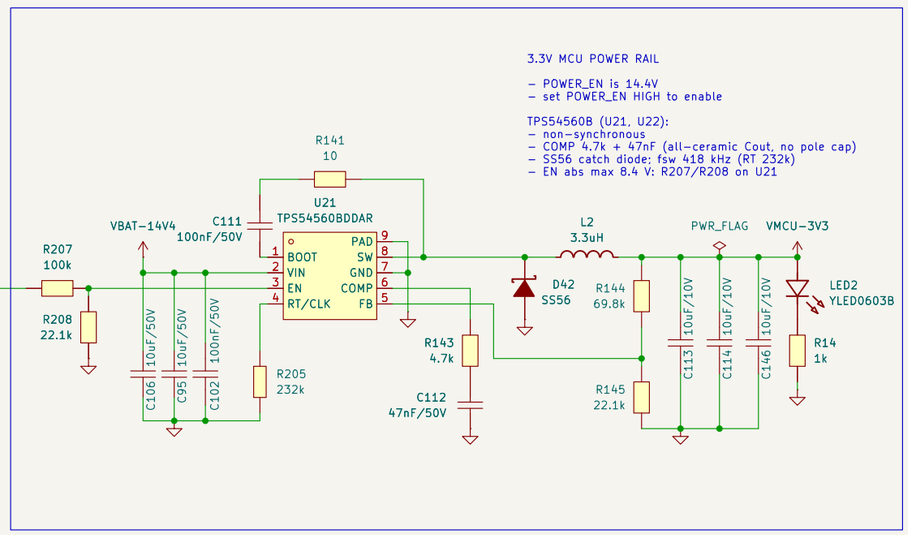
Size: A3 | Date: | Rev:
KiCad E.D.A. 10.0.4 | Id: 1/16

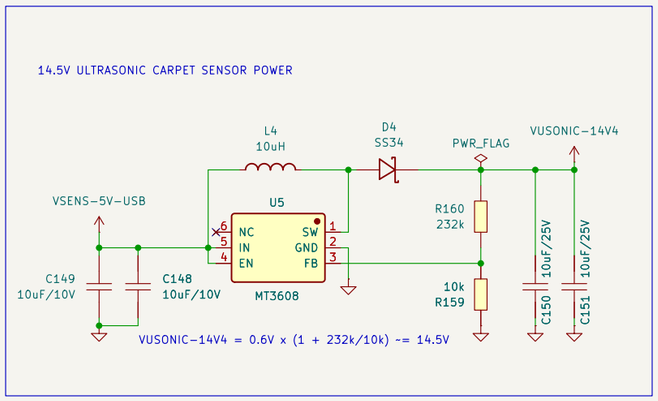


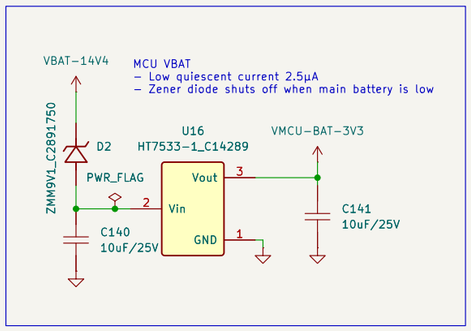










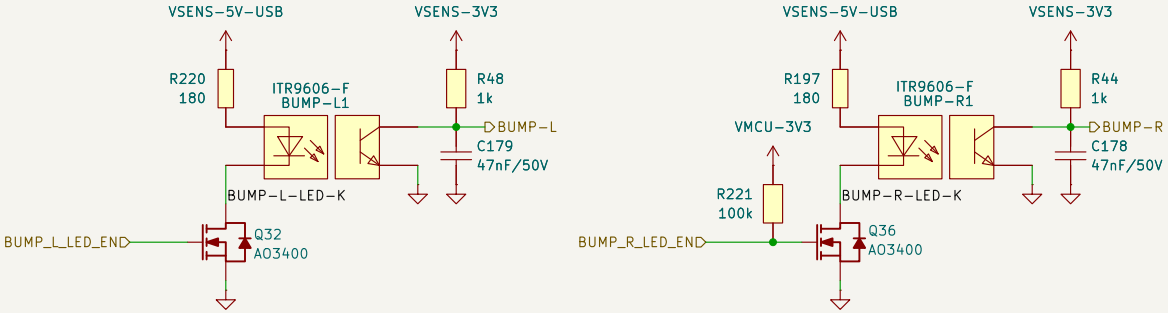


MP1 MP2 MP3 MP4 MP5 MP6

Sheet: /POWER/
File: SYSTEM–POWER.kicad_sch

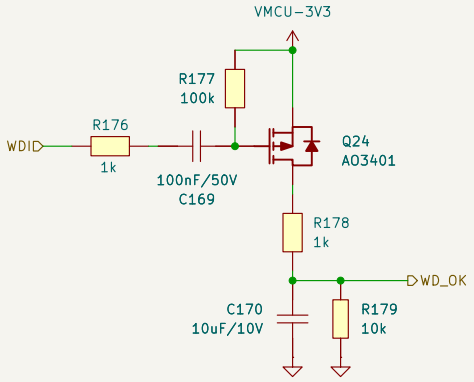
Title:

Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 2/16



- slotted photo-interrupters
- BUMP-X reads low when light passes the slot
- drive BUMP_L/R_LED_EN PWM <<15 usec period to enable
- tune R44, R48, LED PWM duty to let ADC sense full open/close range
  1k keeps photo transistor output out of saturation
  so PA4/PA5 (ADC2) see the transition (ITR9606 IC(ON) 0.5-10 mA)
- R221 enables MCU USB-C Rd on both CC1 and CC2 during MCU *reset*,
  so the phone detects the robot whichever way the cable is plugged in

Sheet: /Bumper Sensors/ File: bumper_sensors.kicad_sch		
Title:		
Size: USLetter	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 3/16



RC CHARGE-PUMP WATCHDOG

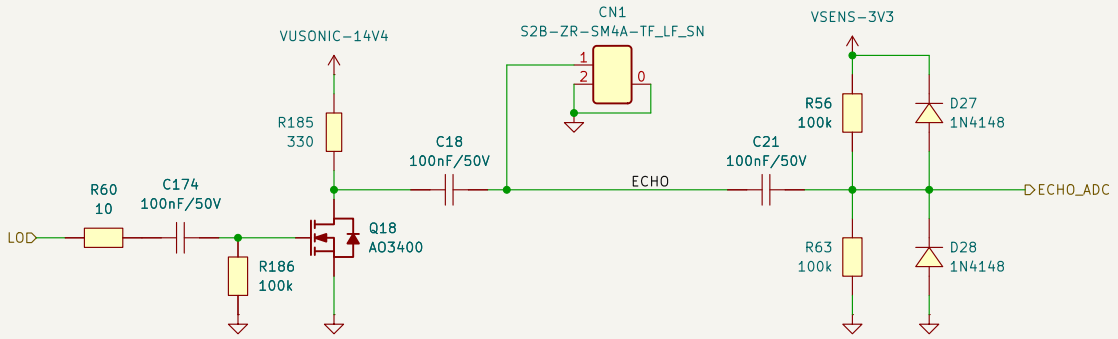
- WD_OK active HIGH
- WDI HIGH->LOW transition pets the dog

Firmware toggles WDI in software (50-1000 Hz, ~50% duty) only while the control loop is healthy.
Never drive WDI from a timer/PWM: a timer keeps running after the CPU hangs.
Each falling edge on WDI pulls the Q24 gate low through C169 and tops up C170 via R178.
WD_OK ~2.7 V while toggling. WDI stuck high, low or floating: <1 V in ~100 ms, <0.4 V in ~190 ms
(10 uF nominal; DC-bias derating shortens this). WD_OK starts at 0 V at power-up (fail-safe).
WD_OK drives N-FET gates only (14.4 V rail switch, LiDAR incl. laser): no resistive loads.
WDI pin: high-Z at reset, no pull-down.

Sheet: /WATCHDOG/	
File: WATCHDOG.kicad_sch	
Title:	
Size: USLetter	Date:
KiCad E.D.A. 10.0.4	Id: 5/16

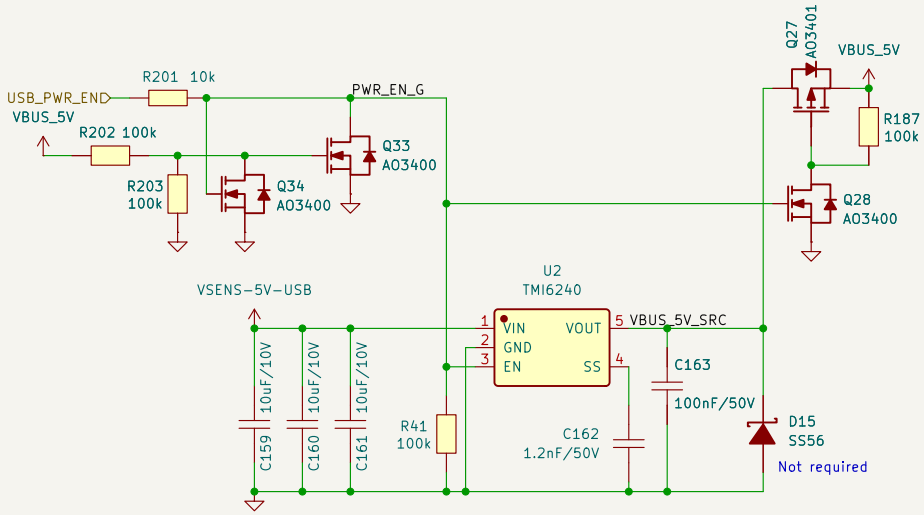
Carpet sensor driver: one low-side AO3400 + 330R pull-up from VUSONIC-14V4 (14.4 V). The P-FET half-bridge was removed: its 100k gate pull-up could not turn it off at 290 kHz (shoot-through). R186 holds LO low in reset/flash.
Drive LO in short bursts only: LO stuck high puts 0.63 W into R185.

C174 AC-couples LO to the Q18 gate: if LO sticks high, the gate decays through R186 (~10 ms) and Q18 turns off, so R185 is never stressed by firmware holding LO high.



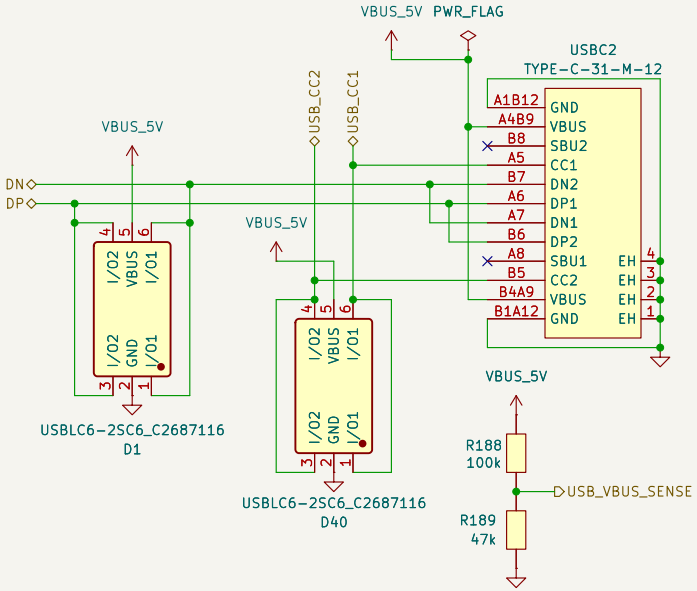
Sheet: /CARPET-SENSOR/		
File: Carpet-sensor.kicad_sch		
Title:		
Size: USLetter	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 6/16

USB-C for SMARTPHONE



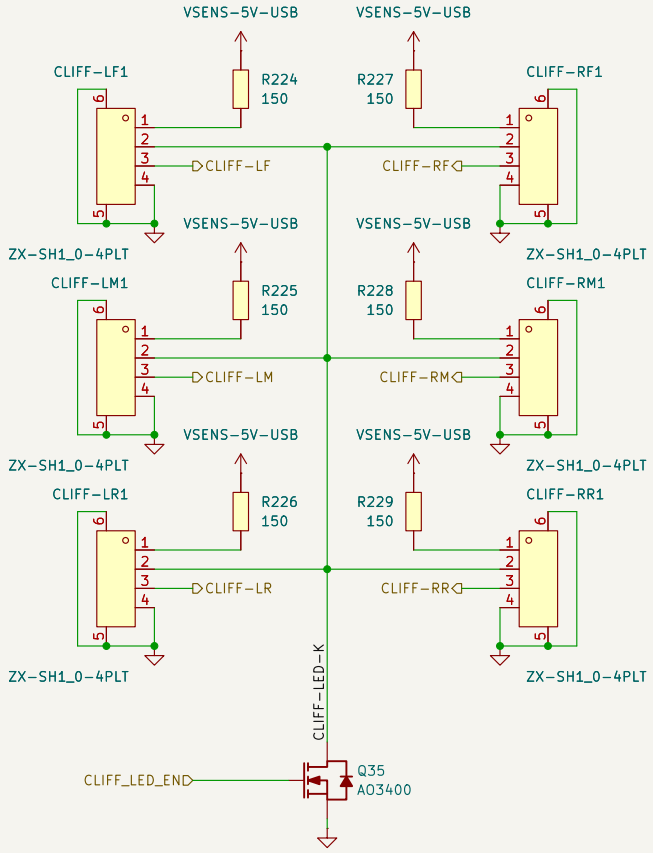
VBUS_5V lockout: the robot 5 V (TMI6240 + Q27) can only turn on while VBUS_5V is ~0 V. Q33 holds PWR_EN_G low if a phone already sources VBUS_5V; once on, Q34 latches it.

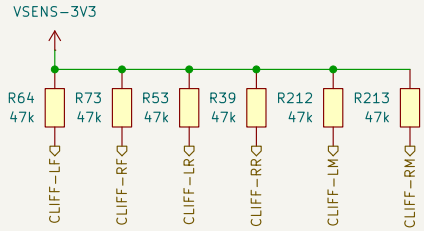
PD power-role swap lets the robot charge the phone while it stays host. Q27 blocks the phone's VBUS_5V from back-feeding the robot 5 V (the TMI6240 body diode conducts VOUT->VIN); USB_PWR_EN turns on TMI6240 and Q27 together.
TMI6240 now runs from VSENS-5V-USB (U25 buck removed).



USB-C (phone): phone is the USB host. CC1/CC2 go to the STM32 UCPD (PB6/PB4): Rd by default.

Sheet: /USB Power/	
File: usb_power.kicad_sch	
Title:	
Size: USLetter | Date:	**Rev:**
KiCad E.D.A. 10.0.4	Id: 7/16



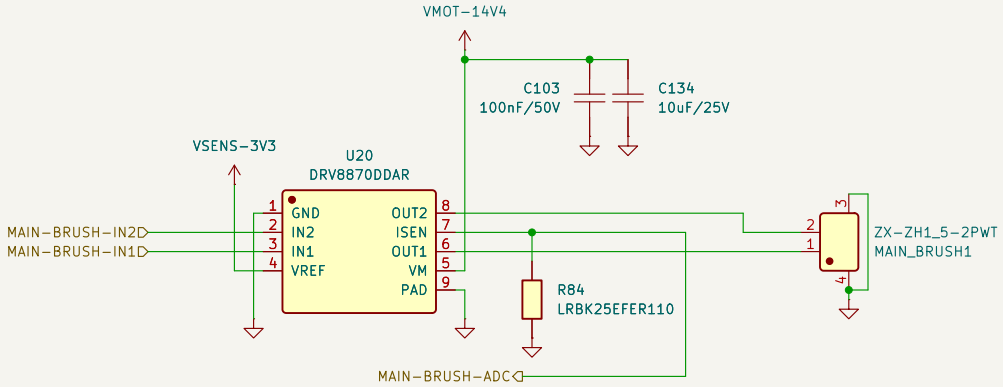


- kicad/cliff-sensor mating boards
- SH 4-pin vertical; use same (pin 1 to pin 1) cable
- set CLIFF_LED_EN LOW to disable
- modulate CLIFF_LED_EN, read each channel LED-on minus LED-off to cancel ambient light
- possible TODO - cliff LEDs 2x or 3x in series, save resistors
  - worsens power rail noise rejection
  - ADC fw could reject noise by correlating the 6 readings
- 0.76W power consumption when enabled continuously
  - drive at ~10% duty cycle to save power
- tweak output pullup
  - a low resistor value can saturating the photo transistor
  - 47K pullup makes for 1ms output settling time
  - can use 10k for faster settling, to reduce duty cycle further
- combine low duty with fast duty LED cycling
  - fast (~10x photo transistor settling time) reduces LED average radiance to avoid photo transistor saturation
- check sensor faults - is output 3.3V or floating?
  - normal no-cliff condition - photo transisstor gets LED light
  Vary LED current and check photo-transistor voltage.

Sheet: /CLIFF/
File: CLIFF.kicad_sch

Title:

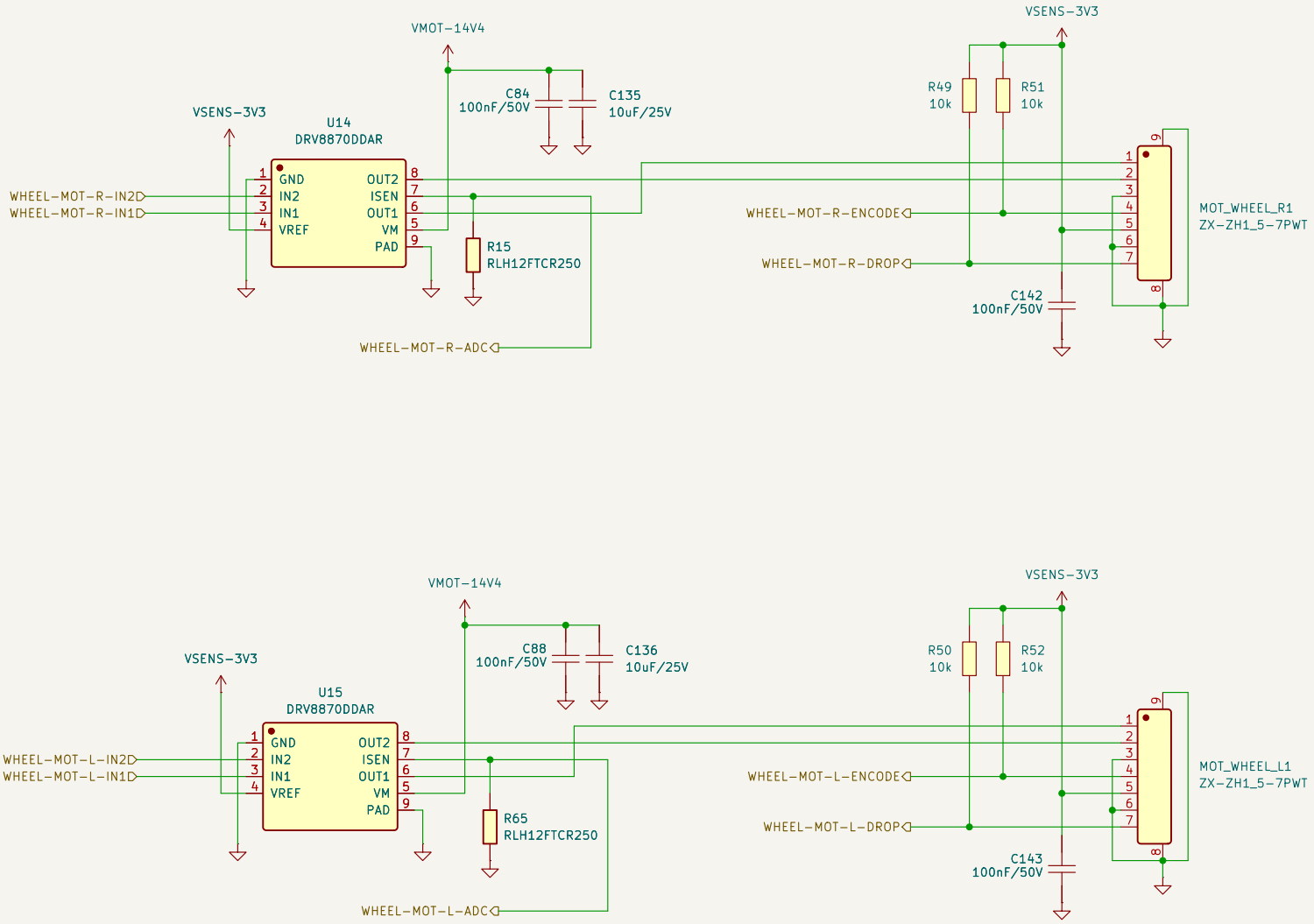
Size: USLetter | Date: | **Rev:**
KiCad E.D.A. 10.0.4 | Id: 19/16



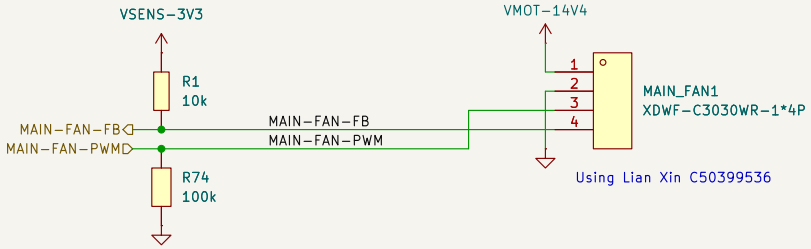
Sheet: /MAIN-BRUSH MOTOR/
File: MAIN-BRUSH-MOTOR .kicad_sch

Title:

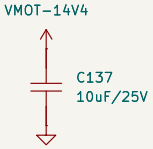
Size: USLetter | Date: | **Rev:**
KiCad E.D.A. 10.0.4 | Id: 22/16



Sheet: /WHEEL MOTOR/	
File: WHEEL-MOTORs .kicad_sch	
Title:	
Size: USLetter | Date:	**Rev:**
KiCad E.D.A. 10.0.4	Id: 23/16



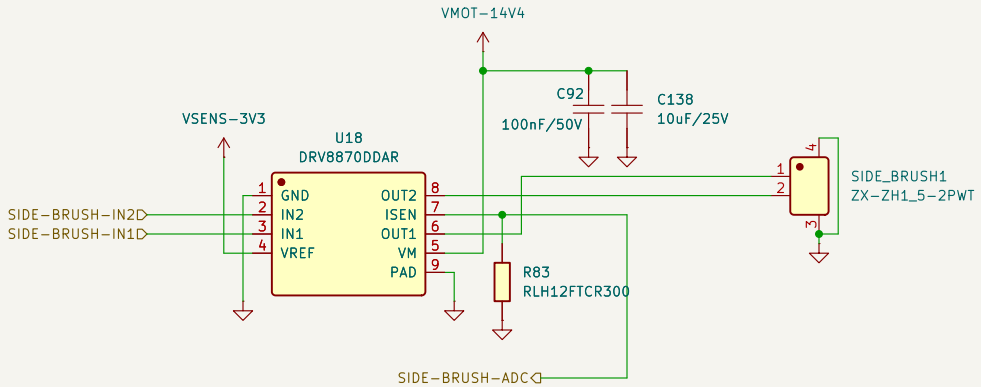
- FB feedback signal is open collector
- set PWM LOW to turn fan OFF
- MAIN_FAN1 matches Dreame L60 PE Ultra PCB



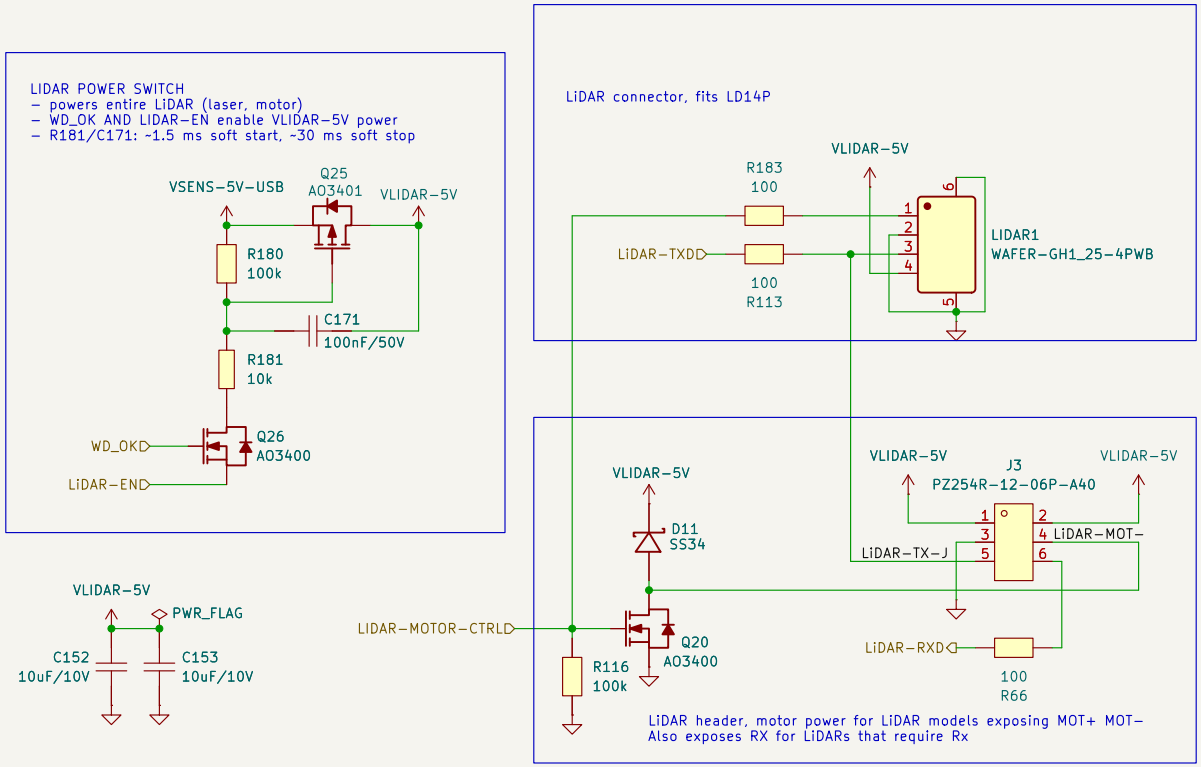
Sheet: /MAIN-FAN/
File: MAIN-FAN .kicad_sch

Title:

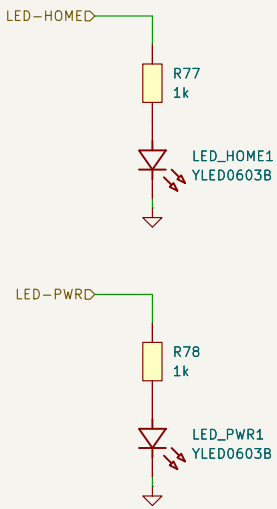
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KiCad E.D.A. 10.0.4		Id: 24/16

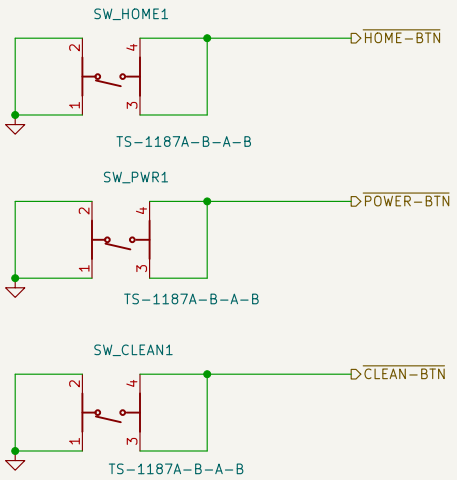


Sheet: /SIDE-BRUSH/	
File: SIDE-BRUSH-MOTORs .kicad_sch	
Title:	
Size: USLetter | Date:	**Rev:**
KiCad E.D.A. 10.0.4	Id: 25/16

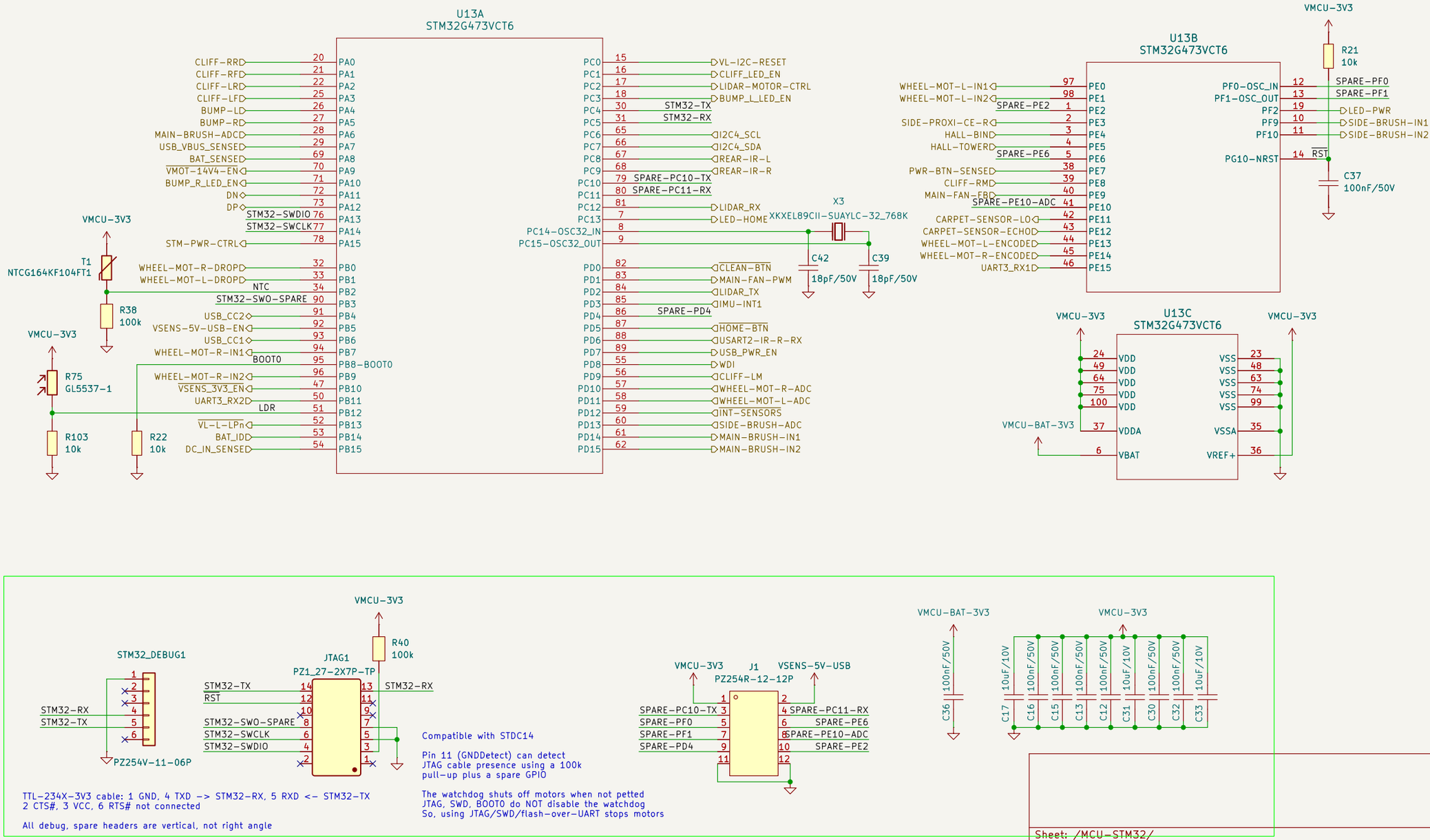


Sheet: /LIDAR/	
File: LIDAR.kicad_sch	
Title:	
Size: USLetter | Date:	**Rev:**
KiCad E.D.A. 10.0.4	Id: 26/16





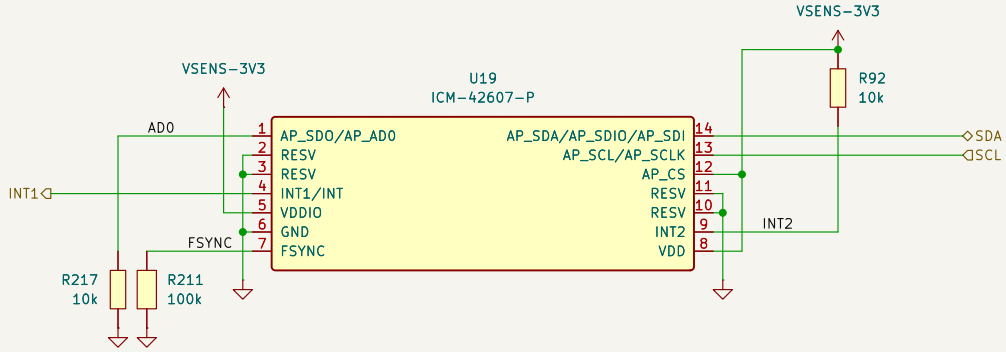
Sheet: /BUTTON-LED/	
File: BUTTON-LEDs .kicad_sch	
Title:	
Size: USLetter | Date:	**Rev:**
KiCad E.D.A. 10.0.4	Id: 28/16



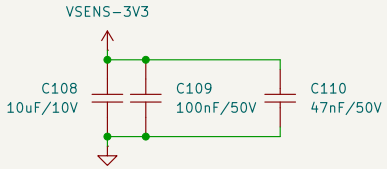
Sheet: /MCU-STM32/
File: STM32G473.kicad_sch

Title:

Size: USLegal	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 30/16



IMU on I2C4 (2.2k pull-ups R5/R6 on the top sheet). AP_AD0 pulled low by R217 (I2C address 0x68).
The IMU is leadless: probe AD0 on R217, INT2 on its pull-up R92 and FSYNC on R211. Run a wire to a spare pin if INT2 or FSYNC is needed.

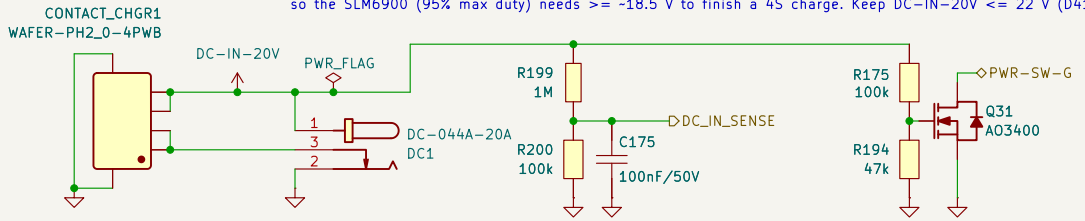


Sheet: /IMU/
File: IMU-ICM-4267-P .kicad_sch

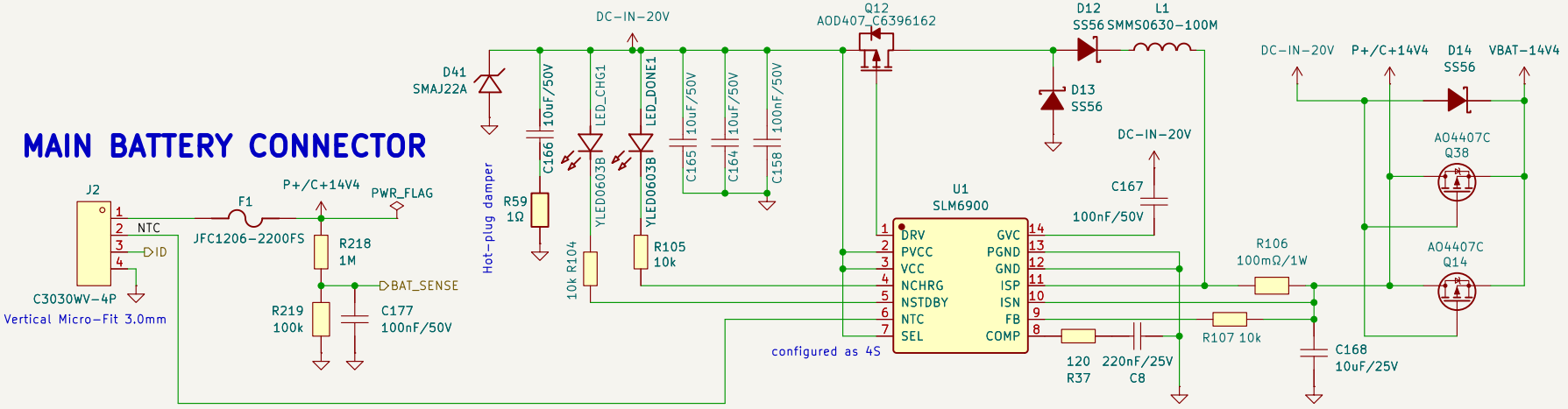
Title:

Size: USLetter | Date: | **Rev:**
KiCad E.D.A. 10.0.4 | Id: 32/16

DC-IN-20V: 20 V brick or dock. With the battery disconnected (Q14 off) VBAT-14V4 = DC-IN-20V - 0.4 V,
so the SLM6900 (95% max duty) needs >= ~18.5 V to finish a 4S charge. Keep DC-IN-20V <= 22 V (D41, Q13 Vgs).



MAIN BATTERY CONNECTOR



Rail-hold: charger power (dock or barrel jack)
turns the MCU rail on and keeps it on.
DC-IN-20V to GND ~130k (R175+R194 || R199+R200)
= dock presence signature.

R106 sets charging current
100 mOhm 1.2 A

Sheet: /BATTERY-CHARGER/	
File: Battery-Charger.kicad_sch	
Title:	
Size: USLetter | Date:	**Rev:**
KiCad E.D.A. 10.0.4	Id: 33/16